Duality AI’s Offroad Semantic Scene Segmentation

Team Name: Anant (Yesh, Swapnil, Shreya, Priya)

Team Code: 5554132f

Tagline: Enhancing semantic segmentation accuracy in synthetic desert environments through multi-model deep learning architectures.

1. Introduction

Autonomous navigation systems operating in off-road and unstructured environments require highly reliable perception systems capable of understanding complex terrain structures. Unlike structured urban environments that rely on predictable road markings and defined boundaries, off-road autonomous navigation requires a nuanced understanding of irregular topographies. Accurate semantic segmentation is the foundational element that enables autonomous Unmanned Ground Vehicles (UGVs) to distinguish between traversable terrain classes and critical obstacles such as vegetation, rocks, logs, and ground clutter. This pixel-level scene understanding directly informs path planning algorithms, significantly improving navigation safety and operational efficiency.

However, collecting and annotating real-world off-road datasets at the pixel level is an incredibly expensive, resource-intensive, and time-consuming process. Synthetic datasets generated through digital twin simulation platforms provide an innovative and efficient alternative. By simulating diverse lighting, weather, and geographical conditions, these platforms allow large volumes of perfectly labeled data to be generated under controlled environmental parameters.

In this project, a comprehensive semantic segmentation system was developed using a synthetic desert dataset. To thoroughly evaluate performance across different computer vision paradigms, three distinct deep learning architectures were trained and evaluated: SegFormer, DeepLabV3 + ResNet50, and UNet++ + TU-ResNet34.

The models were rigorously evaluated using industry-standard metrics, including Intersection over Union (IoU), Precision, Recall, and Confusion Matrix analysis. By comparing these architectures, this study identifies their individual strengths, pinpoints specific failure cases in off-road synthetic data, and proposes optimal deployment scenarios based on varying system requirements.

2. Literature Review

2.1 Duality AI and Digital Twin Simulations

Duality AI is a pioneering technology company that specializes in developing synthetic datasets utilizing highly accurate digital twin simulations. These virtual environments mathematically replicate real-world physical and photometric conditions, allowing complex machine learning models to be trained safely and efficiently without the logistical bottlenecks of physical data collection. Duality AI's core simulation platform, Falcon, is engineered to generate high-fidelity synthetic data specifically tailored for advanced applications such as autonomous navigation, advanced robotics, and defence systems.

2.2 Current Industry Projects and Applications

Project / Application	**Relevance to Semantic Segmentation**
DARPA RACER Program	Focuses on autonomous vehicles navigating complex, off-road environments at high speeds. Requires perception systems that can instantaneously process and segment unknown terrain.
U.S. Army AI Defense	Duality AI contributed synthetic datasets for training state-of-the-art AI models (like YOLOv8) for drone detection. Recognized at the GVSETS 2025 Best Paper awards.

3. Models Used

To ensure a robust evaluation, we selected three models representing different paradigms of computer vision.

Architecture	Backbone	Primary Advantages	Known Limitations
UNet++	TU-ResNet34	Excels at multi-scale feature extraction; produces highly accurate boundaries via dense skip connections; deep supervision flexibility.	Higher computational complexity; slower training times; demands substantial GPU memory allocation.
SegFormer	mit-b2	Transformer-based; strong global contextual understanding of the scene; highly efficient and generalizes well across environments.	Requires meticulous hyperparameter tuning; demands higher initial memory footprints for large attention matrices.
DeepLabV3+	ResNet50	Exceptional multi-scale feature extraction via Atrous Spatial Pyramid Pooling (ASPP); proven, highly stable industrial performance.	Can yield slightly lower accuracy on the finest pixel-level boundaries of highly textured off-road classes.

4. Methodology

4.1 Dataset Preparation and Structure

The dataset consists of high-fidelity synthetic desert environment images (digital twins). Each RGB image is accompanied by a pixel-level mask categorizing the environment into 10 distinct terrain classes: Background, Trees, Lush Bushes, Dry Grass, Dry Bushes, Ground Clutter, Logs, Rocks, Landscape, and Sky.

Dataset Partition	Total Images & Masks	Purpose in Pipeline
Training Data	2,857	Primary data used for model optimization and backpropagation.
Validation Data	317	Unseen during training; utilized to monitor generalization and calculate early stopping metrics.
Testing Data	1,002	Strictly reserved for final inference benchmarking and submission generation.

4.2 Data Preprocessing and Augmentation Pipeline

To enhance model robustness and simulate the dynamic environmental variations a physical UGV camera would experience, a comprehensive pipeline was implemented:

- Preprocessing: Image normalization (scaling pixel values) and image resizing for uniform input dimensions.
- Data Augmentation: Random flipping and rotation simulated the varying pitch, roll, and yaw of a UGV. Photometric adjustments (brightness modifications) mimicked different times of day and shadow casting.

5. Training Strategy and Hardware Configuration

Due to the strict time limitations inherent to the hackathon environment, a parallelized training strategy was deployed. The three models were trained simultaneously on three distinct laptops for 50 epochs utilizing a batch size of 4, allowing for simultaneous benchmarking.

Assigned System	GPU Specification	System RAM	Assigned Architecture
Acer Predator	NVIDIA RTX 4070 (8GB)	16GB	UNet++ (Most computationally heavy)
Lenovo LOQ	NVIDIA RTX 4050 (6GB)	24GB	SegFormer (Memory-intensive Transformer)
ASUS TUF	NVIDIA RTX 4050 (6GB)	16GB	DeepLabV3+ (Stable baseline CNN)

6. Results and Performance Analysis

Following the parallelized training phase, the three models were rigorously evaluated on the strictly unseen validation dataset. This ensured that the models were tested on their true generalization capabilities rather than their ability to memorize the training data. The primary evaluation metrics utilized were Mean Intersection over Union (mIoU), Per-Class Precision, Recall, and comprehensive Confusion Matrix analysis to pinpoint specific areas of overlapping texture bleed.

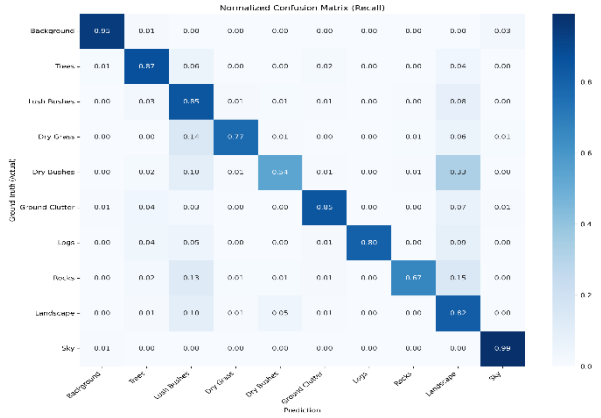
Among the three evaluated architectures, UNet++ achieved the highest mean IoU score, demonstrating clearly superior segmentation capability across the complex synthetic desert terrain. Below is the detailed breakdown of each model's performance.

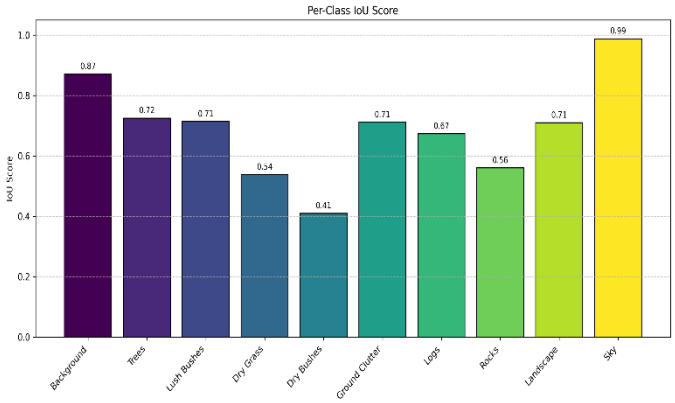
6.1 DeepLabV3+ (ResNet50) Analysis

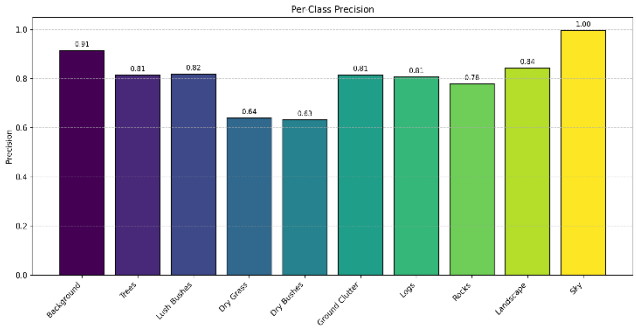
Overall Mean IoU: 0.6899

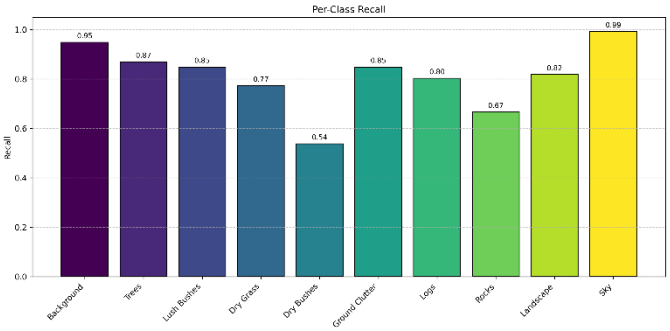
DeepLabV3+ provided highly stable, but occasionally generalized, segmentation performance, closing out the benchmarking with an mIoU of 0.6899. The model utilized Atrous Spatial Pyramid Pooling (ASPP) to perform exceptionally well in detecting dominant terrain regions. It achieved strong, reliable IoU values for the largest classes in the dataset, specifically Background (0.87), Trees (0.72), and Landscape (0.71).

However, empirical analysis of the confusion matrix revealed specific failure cases inherent to traditional CNNs on synthetic data. The model showed notably lower IoU for "Dry Bushes" (0.41) and "Dry Grass" (0.54). This degradation in performance is directly attributed to the model struggling to mathematically separate the severe visual and textural similarities between these overlapping vegetation classes. While it struggled slightly with the finest boundaries, DeepLabV3+ remained a highly reliable baseline architecture.







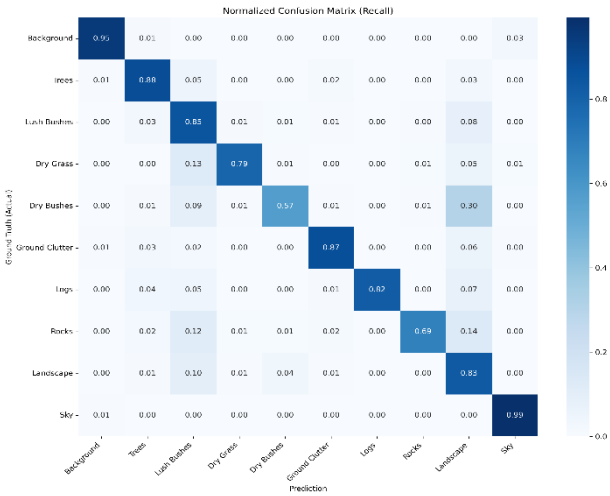


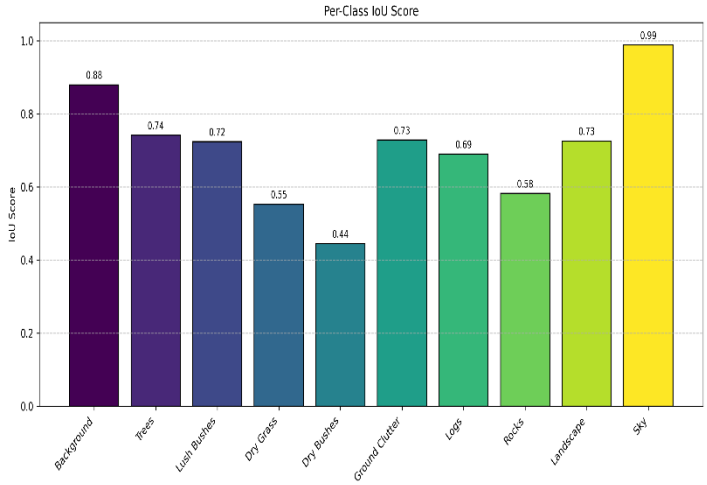
6.2 SegFormer (mit-b2) Analysis

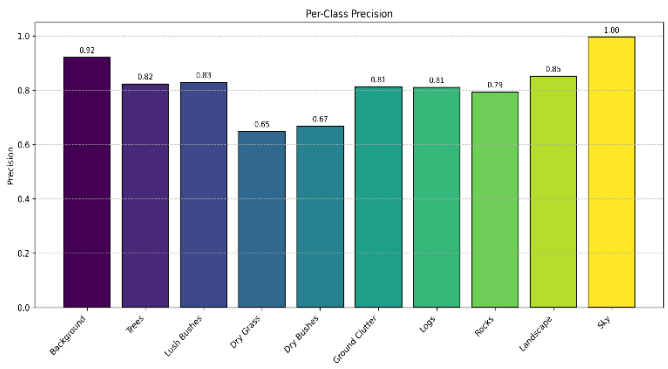
Overall Mean IoU: 0.7056

SegFormer achieved a highly competitive mean IoU of 0.7056, acting as a strong runner-up and demonstrating the profound impact of attention mechanisms on contextual scene understanding. By leveraging its global receptive field, the Transformer-based model excelled in identifying massive, continuous terrain features such as Trees, Ground Clutter, and Landscape.

High precision values for the Background and Sky classes further validated its ability to make accurate predictions with minimal false positives. While the confusion matrix indicated minor overlapping classifications between "Dry Grass" and "Dry Bushes"—an expected outcome given their nearly identical synthetic texture patterns—SegFormer consistently demonstrated a balanced, robust performance with exceptionally strong global awareness of the off-road environment.



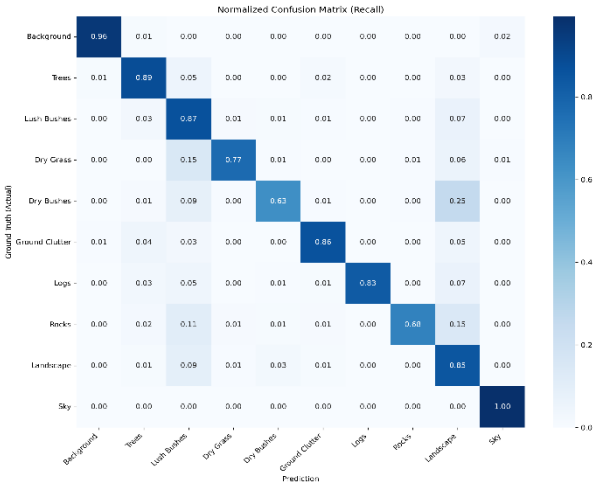


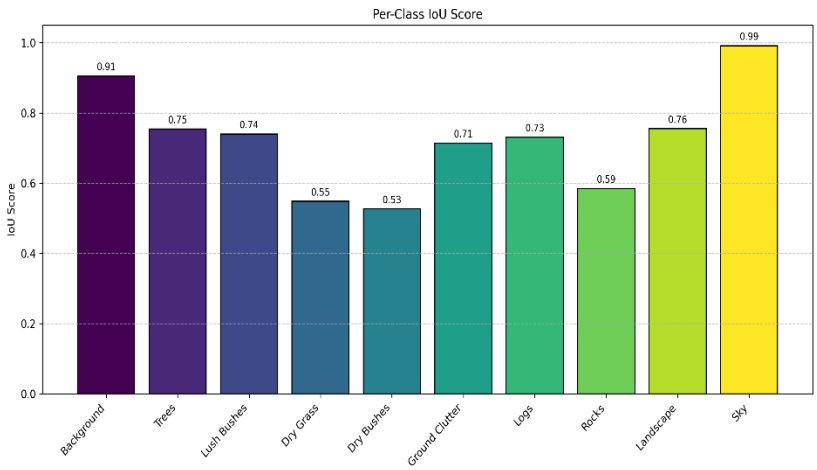


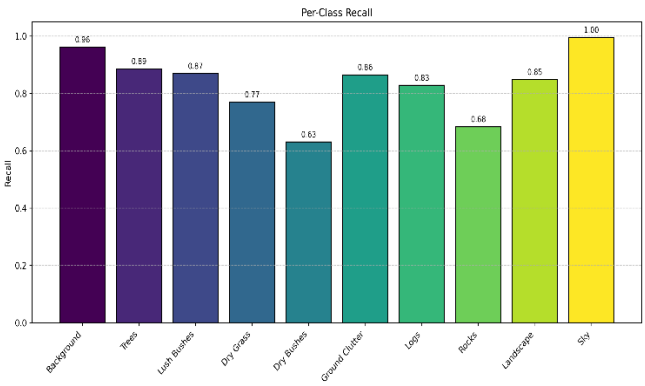
6.3 UNet++ (TU-ResNet34) Analysis - The Winning Architecture

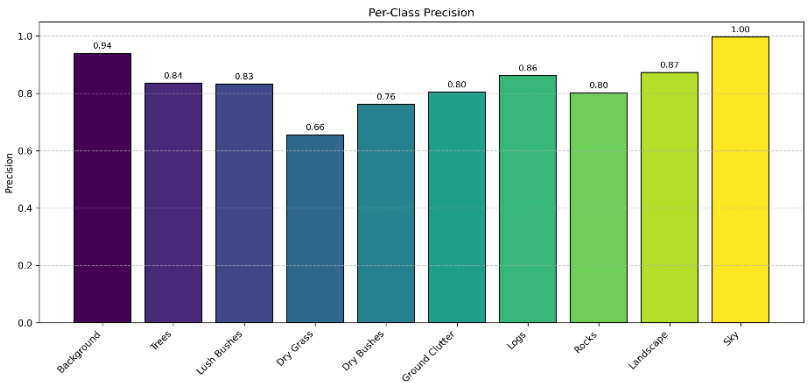
Overall Mean IoU: 0.7253

The UNet++ architecture emerged as the dominant model in this evaluation, achieving the highest overall mIoU score of 0.7253. The integration of nested, dense skip connections allowed the model to achieve far superior feature fusion compared to the baseline models. This architectural advantage directly resulted in highly improved, pixel-perfect boundary detection.The model performed exceptionally well in segmenting critical terrain classes such as Background, Trees, Logs, and Landscape. Precision and recall values remained consistently high across the majority of classes, indicating highly reliable and mathematically accurate segmentation predictions. Most importantly for off-road UGV autonomy, UNet++ consistently produced the most detailed, sharp segmentation boundaries around small, critical obstacles—proving its superior capability in localized feature extraction.









7. Model Functionalities and Requirement-Based Selection

Although UNet++ achieved the highest raw segmentation accuracy, the optimal model selection ultimately depends entirely on the specific edge-compute requirements of the UGV deployment scenario.

UGV System Requirement	Recommended Model	Technical Justification
Highest Segmentation Accuracy	UNet++ + TU-ResNet34	Provides the best IoU, boundary detection, and small obstacle mapping via dense feature fusion.
Real-Time / Lightweight Edge Compute	SegFormer	Highly efficient transformer architecture that balances strong contextual understanding with low computational cost.
Proven Industrial Stability	DeepLabV3+ + ResNet50	A heavily researched, reliable baseline providing stable segmentation results across diverse conditions.

8. Engineering Challenges and Solutions

Throughout the rapid development lifecycle of this hackathon, several significant engineering challenges were identified and mitigated.

Engineering Challenge	Impact / Cause	Deployed Solution
1. High Computational Complexity	Training deep learning models on large, high-resolution images caused massive computational bottlenecks and slow epochs.	Implemented a highly optimized TU-ResNet34 backbone for UNet++ to drastically reduce complexity while maintaining accuracy.
2. Hackathon Time Constraints	Sequential training was mathematically impossible within the hackathon deadline.	Engineered a distributed, parallelized workflow, deploying all three architectures simultaneously across three independent laptops.

Engineering Challenge	Impact / Cause	Deployed Solution
3. Severe Class Similarity	"Dry Bushes" and "Dry Grass" shared nearly identical, overlapping synthetic textures, causing confusion.	Utilized fundamentally different architectures (CNNs vs. Transformers) to capture entirely different mathematical feature representations.
4. The Sim-to-Real Gap	Models trained on synthetic twins frequently fail to generalize in real-world physical deployments.	Built an aggressive, on-the-fly data augmentation pipeline (rotations, brightness shifts) to force high model robustness.

9. Conclusion

This project successfully developed and benchmarked a highly accurate semantic segmentation system tailored specifically for off-road terrain analysis using synthetic digital twin datasets. By deploying a parallelized training strategy across three separate laptops, rapid experimentation was achieved under strict hackathon time constraints.

Three state-of-the-art models were rigorously evaluated. Empirical analysis proved that UNet++ achieved the highest overall mean IoU score (0.7253), demonstrating vastly superior segmentation performance and boundary detection. By thoroughly analyzing these architectures, this study provides a comprehensive framework for deploying robust perception systems based on varying computational and accuracy requirements for autonomous off-road navigation.

10. Future Work

To further bridge the sim-to-real gap and enhance the capabilities of autonomous perception systems, future iterations of this project will explore:

1. Training with Larger Datasets: Expanding digital twin simulations to include varying weather conditions (rain, fog) and significantly larger datasets.
2. Exploring Additional Architectures: Benchmarking newer, highly efficient models designed specifically for mobile edge computing.
3. Improving Ensemble Methods: Developing advanced, weighted-average ensemble scripts to dynamically combine the strengths of Transformers and CNNs in real-time.
4. Optimizing for Real-Time Deployment: Exporting the model weights into optimized inference engines (such as TensorRT) to ensure ultra-low latency, real-time segmentation deployment on physical UGV platforms.